

Anurag Gupta

+91 9406642990 | anurag210103@gmail.com | linkedin.com/anuraggupta77 | github.com/a-n-u-r-a-g | Portfolio

EDUCATION

Vellore Institute of Technology, Bhopal

M.Tech (Integrated) - Computer Science and Engineering with Specialization in Cyber Security

CGPA: 8.98

Oct. 2020 - Jun. 2025

D.A.V. Public School

Singrauli, Madhya Pradesh

XII: 74.6%

Apr. 2019 - May 2020

D.A.V. Public School

Singrauli, Madhya Pradesh

X: 76.4%

Apr. 2017 - May 2018

TECHNICAL SKILLS

Languages: Python, Java, C#, PHP, SQL, HTML, CSS, JavaScript

Frameworks/Suits: React JS, Node JS, MongoDB, REST APIs, SAP, GRC, NIST, Agile Methodologies,

Tools: Power BI, Tableau, Selenium, Docker, Kubernetes, Git, Jira, Google Colab, Windows, Linux/Unix

Familiar Skills: Operating System, Networking, DBMS, Cloud Computing (AWS), Problem Solving, Data Structure, Data Analysis/Analytics, Reporting & Documentations, Team Work, SDLC

PROJECTS

Insight-Pro: eCommerce Analytics Suite

Python, SQL, Power BI, Google Analytics, AWS, Excel

Oct. 2024 - Nov. 2024

- Engineered optimized SQL and Snowflake queries that **improved data retrieval speed by 45%** and increased reporting accuracy by **30%** for customer behavior, product performance, and sales trend analysis.
- Built automated ETL workflows to process and transform business data, **cutting manual processing time by 60%** and boosting overall operational efficiency by **50%**.
- Designed interactive Power BI dashboards that **reduced reporting turnaround time by 70%**, enabling stakeholders to monitor KPIs and make data-driven decisions more effectively.
- Established data validation and quality control processes that **reduced data inconsistencies by 40%** while maintaining **99.5% data accuracy** across business and financial reports.

Adaptive Hybrid Ensemble Learning with Explainable AI for Anomaly Detection

Python, Scikit-learn, Machine Learning, Deep Learning

Aug. 2024 - Oct. 2024

- Designed and implemented a hybrid ensemble model combining CNN, Bi-LSTM, Random Forest, Gradient Boost, and XG Boost to enhance anomaly detection accuracy in network traffic.
- Integrated SHAP-based Explainable AI techniques to improve model transparency and interpretability, supporting critical decision-making in cybersecurity operations.
- Developed a scalable pipeline for preprocessing, multi-model classification, and real-time explainability, tailored for high-volume and heterogeneous network environments.
- Achieved 99.98% accuracy on CICIDS2017 and 97.71% on UNSW-NB15 with false positive rates as low as 0.02% and 0.01%, validated using ROC analysis and SHAP visualizations.

Container Identity and Access Management

Python, Next JS, JSON, Docker, Bash Scripting

Aug. 2023 - Oct. 2023

- Designed and implemented a secure container identity framework by generating hashed identifiers (Docker IDs, container IDs) linked to access privileges, ensuring granular identity verification in runtime and testing environments.
- Developed a Python-based logic layer integrated with Docker and Bash to automate identity generation, access mapping, and secure role-based controls across containerized services.
- Engineered a modular front-end using Next.js for real-time visibility into container access rights and lifecycle events, enhancing security posture and operational transparency.
- Improved container runtime security, isolation, and auditability in DevSecOps pipelines by enforcing dynamic identity binding—resulting in a 40% reduction in unauthorized access attempts and a 30% improvement in audit traceability.

CERTIFICATIONS

- **AWS Certified Cloud Practitioner** | Amazon Web Service
- **The Bits and Bytes of Computer Networking by Google** | Coursera
- **Junior Cybersecurity Analyst Career Path** | Cisco

Dec. 2023

Apr. 2023

Apr. 2022

ACHIEVEMENTS/ CO-CURRICULAR

- Published a paper entitled "[Hybrid Ensemble Learning with Explainable AI for Anomaly Detection in Network Traffic](#)"
- Earned 5 Stars in Java, Python, and C/C++ on Hackerrank.
- Solved 200+ problems on Leetcode and 300+ problems on Geeks for Geeks.
- Certified Acquiring Data by NASSCOM & Skill India.

LEADERSHIP / EXTRACURRICULAR

Sports Selection – AdvITYa

- Coordinated and supervised for a smooth and fair competition during AdvITYa'23

Executive Coordinator, Techno-Mech Club

- Directed club activities, orchestrated events, and enhanced member engagement through various technical and mechanical projects.

LANGUAGES AND INTERESTS

Languages: English, Hindi **Interests:** Market Analysis (Fintech), Geopolitics, Badminton, Chess, Podcast